

Digital Forensics and Incident Response (DFIR) Report

Date of Report: Friday, October 2, 2026, 16:07:42

Subject File: evidence03.pcap

Analysis Scope: Packet Ranges 1168.1.10 – 891 and 862000 – 1778

Status: Closed - Benign Activity

1. Executive Summary

A forensic analysis of `evidence03.pcap` was conducted to identify potential indicators of compromise (IoCs), unauthorized data exfiltration, or malicious network activity. The analysis focused on two distinct packet ranges involving internal host `192.168.1.10`. The findings indicate standard, automated application behavior consistent with a user interacting with Apple-integrated media services (e.g., iTunes, Apple TV). No evidence of malicious intent, unauthorized lateral movement, or data breaches was identified.

2. Incident Overview

The captured traffic is characterized by outbound-oriented HTTP/DNS activity. The primary activities observed include:

- * **Media Asset Retrieval:** High-volume downloading of JPEG image assets (album art/thumbnails) from Apple/Akamai Content Delivery Networks (CDNs).
- * **Incremental Search Functionality:** A “type-ahead” pattern where the client sends successive API requests to `ax.search.itunes.apple.com` as a search string is incrementally built (e.g., `q=iknow` → `q=iknowyou`).
- * **Telemetry & Analytics:** Simultaneous transmission of search strings to Apple analytics endpoints (`66.235.132.121`) for service optimization and tracking.

3. Technical Analysis

Network Infrastructure & Hosts

Role	IP Address	Description
Internal Source	<code>192.168.1.10</code>	Primary Client (Likely mobile/media device)
External (CDN)	<code>8.18.65.82</code> , <code>8.18.65.88</code> , <code>8.18.65.89</code>	Apple/Akamai Content Delivery Network
External (DNS)	<code>4.2.2.1</code>	Google Public DNS

Role	IP Address	Description
External (Analytics)	66.235.132.121	Apple Analytics/Telemetry Endpoint

Protocol & Traffic Observations

- **Protocols:** HTTP/1.1, DNS, TCP, HTTP/XML.
- **Traffic Pattern:** The traffic follows a predictable, user-driven pattern. The transition from DNS resolution to HTTP GET requests for media assets and search queries aligns with standard application-layer behavior for media streaming services.
- **Connection Management:**
 - Standard TCP three-way handshakes and orderly terminations (FIN, ACK) were observed.
 - **Anomalies:** Minor network instability was noted, including a TCP Retransmission of a FIN, ACK packet and a TCP [RST, ACK] from a remote server, likely due to transient network congestion or session timeouts. These do not indicate malicious activity.

4. Key Findings

Finding	Detail	Severity
Search Behavior	Observed incremental API calls (MZSearch.woa) corresponding to real-time user input.	Informational
Data Integrity	No evidence of unauthorized data exfiltration or unauthorized inbound connections.	Low
Protocol Integrity	Traffic adheres to standard HTTP/DNS/TCP specifications; no protocol anomalies detected.	Low

Finding	Detail	Severity
Service Interaction	All external communications are directed toward verified Apple/Akamai infrastructure.	Low

5. Conclusion

The analysis of `evidence03.pcap` confirms that the observed network traffic is legitimate and consistent with the automated functions of a media-consuming device. The patterns of incremental searching and asset downloading are characteristic of modern UI/UX design in streaming applications. No further investigation is required.